

NEHA RAVISHANKAR RAO

ASIC / FPGA Design & Verification Engineer

+1 (919)717-4063 ◊ nnavish24@gmail.com / nnavish3@charlotte.edu ◊ Charlotte, NC

linkedin.com/in/neharrao ◊ github.com/neharrao ◊ neharrao.github.io

SUMMARY

Master's in Computer Engineering with experience in ASIC/FPGA design and verification using SystemVerilog, UVM, assertions, and coverage-driven methodologies. Skilled in building OOP-based testbenches, verifying protocols (AXI, SPI, I2C, UART, Wishbone), and debugging SoC-level designs.

TECHNICAL SKILLS

Verification Languages: SystemVerilog, Verilog, VHDL

Verification Methodologies: UVM(basics), Constraint-Random Testing, Coverage-Driven Verification, Functional & Code Coverage, SVA, Formal Verification (basics)

FPGA Tools & Flows: FPGA Prototyping, Synthesis, Implementation, Bitstream Generation

Simulation & Debug Tools: Vivado ILA, Vivado, EPWave, Waveform Viewer, ModelSim, QuestaSim

Architecture Knowledge: Memory Subsystems, CPU Architecture, FSMs, Protocols (SPI, UART, I2C, AXI, Wishbone)

Programming & Scripting: Python, C++, Makefile, Git

EDUCATION

Master of Science in Computer Engineering, University of North Carolina, Charlotte August 2023 - August 2025

Relevant coursework: Computer Architecture, Reconfigurable Computing, Intro to VHDL, Advanced Embedded Systems, Data Communication and Networking, Interconnect Networks

GPA: 3.6

Bachelor of Technology in Electronics and Communication Engineering, MITS, India August 2017 - May 2021

Relevant coursework: Logic Circuit Design, Computer Organization, Microprocessors & Microcontrollers, VLSI, Embedded Systems, Digital Communication, Object Oriented Programming

GPA: 3.5

EXPERIENCE

Graduate Teaching Assistant, UNCC, Charlotte, NC January 2024 – May 2025

- Led recitation sessions, graded assignments & tests for courses in Logic Design, Embedded Systems, and C++ Programming.

Technical Consultant (Security Analyst), IBM, India October 2021 – May 2023

- Managed firewall configurations and security policies, ensuring 99.9% uptime.
- Developed monitoring workflows, reducing network troubleshooting time by 30%.

PROJECTS

• SystemVerilog-Based Verification of Memory, Protocol, and Interface IP

- Developed layered SystemVerilog testbenches with Object-Oriented Programming (OOP) constructs, constraint randomization, and inter-process communication for RTL verification.
- Built and verified environments for memory blocks, combinational/sequential designs, and standard protocols (SPI, UART, I2C, APB, AHB, AXI, Wishbone) using coverage-driven testing.

• Dual-Core Processor System on PYNQ Z2

- Built a dual-core processor system with ARM Cortex-A9 and MicroBlaze on PYNQ Z2 for real-time data exchange.
- Verified AXI-based BRAM communication and debugged inter-core sync using Vivado ILA and UART logs.

• SystemVerilog PWM Core Development on PYNQ-Z2 FPGA

- Designed and implemented a configurable-duty-cycle PWM core in SystemVerilog on PYNQ-Z2, supporting 0–100% duty cycle with 8-bit resolution, and verified 1 kHz output via simulation.
- Completed full FPGA flow—coding, synthesis, implementation, bitstream generation, and on-board hardware validation.

• VHDL Microprocessor Design and Simulation

- Designed an 8-bit custom processor with RTL simulation to verify control/data path.
- Used waveform viewer and assertions to debug instruction decode and memory logic errors.

ACCOMPLISHMENTS & LEADERSHIP

• Treasurer, IEEE Women in Engineering Student Chapter Aug 2024 – Present

• Completed 5 years training in Indian Navy as Leading Sea Cadet - Leadership, discipline, and technical precision.